Lecture 29: Course Wrap-Up and Data Science Next Steps

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Recipe of the Day!

Holiday Cocktails

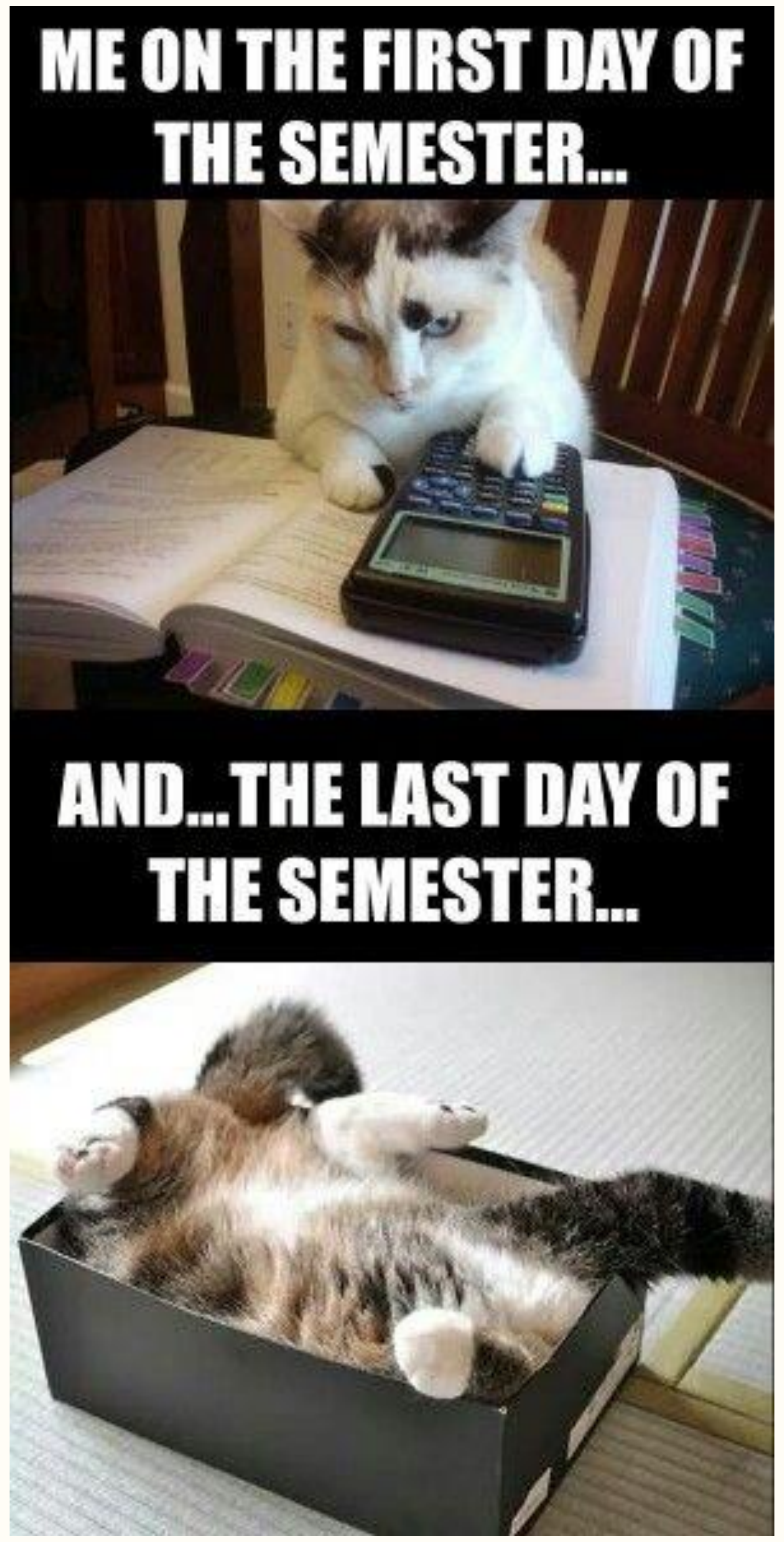


Holiday Mocktails



Announcements

- No lab this week
- Project files due today(!!) by 12pm (noon)



Course Highlight Reel

What did we learn?

The Data Science Pipeline

Collecting and importing (loading) the data

Processing (cleaning, wrangling) the data

Visualizing and summarizing the data

Building models (statistical and ML)

Interpretation and communication of results

Topics Covered

R Basics	GitHub	Data Wrangling	Data Visualization	Data Summarization and Presentation	Linear Regression	Machine Learning
• R Markdown • Data types • Vectors • Sorting • Vector arithmetic • Indexing • Data wrangling • Importing data • Functions • For loops • If-else statements	• Cloning repositories via RStudio • Committing and pushing assignments via RStudio • Pulling course notes via RStudio	• Tidy data format • Importing data • Reshaping data • Join functions and combining tables • Dates and times	• ggplot2 • Data visualization principles • Faceting • Fixing scales • Time series plots • Transformations • Ordering by a value • Making comparisons • Maps	• Numerical summaries • Frequency tables • The tableone package • Reporting missing data • Types of missingness • Simple imputation methods • Advanced imputation methods • Survey skip patterns	• Correlation • Simple linear regression • Evaluation of model assumptions • Multivariate linear regression • Collinearity • Model selection	• Fundamentals of ML including train/test split and the ML process • Cross-validation • Logistic regression, Naive Bayes, kNN, Decision trees, Random Forests, LASSO, Ridge regression, PCA, k-means clustering • ROC and AUROC • Confusion matrix • Performance metrics • Scaling • Imputation

Data Science Next Steps

Courses at Harvard

Harvard Chan

- BST 263 Statistical Learning (R or python)
  - Half theoretical, half applied
- BST 261 Data Science II (python)
  - Deep learning
- BST 221 Data Structures and Algorithms (python)
  - More of a computer science course
- ID 529 Data management and analytic workflows in R
  - January course

Harvard Grad School of Education

- EDU S022 Introduction to statistical computing and data science in education
  - A lot of overlap with BST 219
- EDU S052 Intermediate and Advanced statistical methods for applied educational research (R or Stata)
- EDU S030 Intermediate statistics for educational research: applied linear regression (Stata)

Courses at Harvard

Other Harvard Courses

- APMTH 120: applied linear algebra and big data
  - Linear algebra and R
- APCOMP 221 Critical Thinking in Data Science
  - Data Science ethics
- STAT 102 Introduction to Statistics for Life Sciences

Skills Development

- Keep using R for research or other projects!
- Learn Linear and Matrix Algebra
- Diversify your Programming Languages (Python, SQL, C++, Java, etc.)
- Kaggle competitions
- Online courses on Coursera, edX, Data Camp, etc.
- Data Science Meetups in Boston
- Data science podcasts and social media
  - Everyday AI podcast
- Data science blogs
- Data science tutorials and videos

Become a TF!

We are always in need of teaching fellows for our Biostats courses, including this one!